# Pandelis Zembashis

**CTO | Agent Harness Engineer | AI Coding Runtime, Tools & Sandboxing**

p@ndel.is | pa.ndel.is | github.com/PandelisZ | x.com/PandelisZ | linkedin.com/in/Pandelis

## EXECUTIVE SUMMARY

CTO and hands-on engineer building the core harness for AI coding agents. At Cosine, led the runtime and platform work behind an applied AI lab for software engineering: Go agent harness, tool execution, event-bus orchestration, deterministic replay, multi-agent workflows, repository search, browser use, memory, Kubernetes/gVisor sandboxing, and enterprise deployments. Direct fit for Poolside's Agent Harness role: building reliable, high-performance agent frameworks and secure isolated runtimes that connect research progress to production developer experience.

## PROFESSIONAL EXPERIENCE

**Cosine | CTO | Oct. 2023 – Present**

*Applied AI lab building autonomous coding agents, custom harnesses, secure runtimes, and developer products.*

- Lead product and engineering for Cosine's AI coding platform, operating across agent framework design, runtime implementation, tool orchestration, research feedback loops, and enterprise deployment.
- Built Cosine's own Go harness rather than wrapping another agent loop: streaming tool calls, checkpoints, deterministic replay, memory writes, MCP/LSP integration, subagents, file locks, and structured rollout logs.
- Built sandboxed execution from 2024 onward: Kubernetes-scheduled remote agents, gVisor/runsc isolation, ephemeral per-session environments, network-policy boundaries, and secure shell/tool access for coding agents.
- Shipped one runtime across CLI, desktop, VS Code/WebAssembly, API-server sessions, and remote worker pods, keeping every product surface aligned through one event model and one tool layer.
- Reworked filesystem and repository-ingestion paths around native Go concurrency, bundled ripgrep-class search, streaming file reads, and low-overhead tool results for large enterprise codebases.
- Built the agent evaluation flywheel around the harness: replayable traces, tests, LSP signals, checkpoints, reviews, and parallel sandboxed evaluation runs.
- Supported cloud, private VPC, on-prem, and air-gapped deployments where source code, model artifacts, logs, and tenant data remain isolated.

**11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023**

*Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.*

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled outbound email and event-driven automation systems that processed millions of email chains and billions of tokens.
- Established engineering cadence, incident response habits, and onboarding rhythms during rapid seed-to-Series A growth.

**Papaya | Technical Lead | Feb. 2022 – Mar. 2023**

*Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.*

- Led delivery of a fleet management platform for light electric vehicles in Next.js, React Query, and Tailwind CSS.
- Modernized the Ruby on Rails backend by introducing GraphQL APIs, improving product iteration across web, operations, and marketplace workflows.
- Established development cadences and product intake processes that improved execution across engineering and product.

**Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022**

*Rapid grocery delivery startup building warehouse, ordering, and customer systems.*

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, incident response, and security.

**finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021**

*White-label fintech platform powering digital banking products and apps.*

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

**IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020**
*Cloud IoT platform unifying device protocols behind a GraphQL API.*

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

**Substrakt | Software Developer | Aug. 2018 – Jun. 2019**
*Digital product agency building bespoke publishing, arts, and cultural web platforms.*

- Built bespoke WordPress features, search experiences, and internal tooling for design-led client websites.
- Modernized the JavaScript toolchain from Gulp to Webpack, improving development and deployment workflows.

**D4 Software | Software Engineer | Jun. 2017 – Jul. 2018**
*Product engineering consultancy building client platforms, tools, and cloud infrastructure.*

- Built client-facing features and internal tools across product and platform engagements using JavaScript and .NET.
- Maintained AWS infrastructure, serverless services, databases, and virtual machines while modernizing the frontend stack with Webpack, TypeScript, and Vue.js.

## WRITING & PUBLIC WORK

**Why we built our own harness**: Technical deep dive on Cosine's Go harness, tool execution, replay, multi-agent orchestration, gVisor-sandboxed Kubernetes substrate, and ML feedback loop.
**Cosine Swarm**: Long-horizon agent orchestration, parallel coding work, and coordination patterns for agentic software engineering.
**Cosine CLI runtime**: Single-binary distribution, native search, local workflows, and agent execution improvements.

## EDUCATION

**Birmingham City University** | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

## AREAS OF EXPERTISE

**Agent Harness & Runtime**: Go, Python, event-bus design, tool execution, checkpoints, deterministic replay, subagents, memory systems, MCP/LSP integration
**Sandboxing & Orchestration**: Kubernetes, gVisor/runsc, containers, remote workers, ephemeral environments, shell control, WebSockets, protobuf, network policy
**Performance & Developer Experience**: Repository indexing, filesystem traversal, native search, streaming I/O, CLI tooling, VS Code/WebAssembly, desktop runtimes
**Enterprise AI Platforms**: Cloud, private VPC, on-prem, air-gapped deployment, RBAC/SSO, audit logging, tenant isolation, observability, incident debugging